

TECHNICAL REPORT · ROAD TO GREEN

# Rooftop Solar PV System
# Design Basis & Cost Estimate

Marriott Hotel – Mezzanine Roof – Phase 1

| | |
|---|---|
| Document No. | NILESUN POWER RF002 |
| Revision | B – Updated costing and financials |
| Date | 9 November 2026 |
| Client | Kampala Marriott Hotel |
| System capacity | 185.4 kWp DC / 160 kW AC, grid-tied |

Prepared by NileSun Power Limited
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Document Control

| Rev | Date | Description | Prepared | Checked | Approved |
|---|---|---|---|---|---|
| A | 3/11/2026 | Version A | Mahil Fernando | Prasanna Dissanayake | Manjula Perera |
| B | 9/11/2026 | Updated costing & financials | Mahil Fernando | Prasanna Dissanayake | Manjula Perera |

## Reference documents

| Reference | Title |
|---|---|
| D18102-DD-A-104 | Mezzanine floor plan incl. roof PV zones (architect) |
| PV-LAY-D18102-002 | PV layout with ballast structure, 10° tilt |
| EPJ-26110401 V1.0 | Mounting structure drawings (2P and 1P tables) |

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## 1. Executive Summary

This report sets out the design basis, layout, energy yield and budget cost for a grid-tied rooftop solar photovoltaic (PV) system on the mezzanine-level roof of the Marriott Hotel,

The system has no battery storage and supplies the hotel's own load behind the utility meter.

| 185.4 kWp | 274 MWh | USD 103k | 5.2 yrs |
|---|---|---|---|
| DC capacity (299 × 620 Wp) | Year-1 generation | Total project cost (0.55 USD/Wp) | Simple payback |

### Key findings

From Energy audit and feasibility study key findings.

| Item | Value |
|---|---|
| Roof area (mezzanine roof) | 1,150 m² (Available area for solar installation) |
| Plant capacity | 185.4 kWp |
| Modules | 299 × 620 Wp (13 rows of 23 solar modules) |
| Structure | Fixed-tilt ballast structure, 10° |
| Inverters | 2 × 80 kW (160 kW AC) |
| Hotel consumption | 1,000,000 kWh/month |
| Self-consumption | 100% (no export, no battery) |
| Year-1 generation | 274 MWh (2.3% of annual consumption) |
| Avoided tariff | USD 0.07/kWh |
| Year-1 savings | USD 19,275 |
| Total project cost | USD 102,789 |
| Simple payback | 5.2 years |
| Project IRR | 19.8% |
| NPV (25 years at 10%) | USD 92,606 |
| Cost of solar energy (LCOE) | 0.043 USD/kWh |
| Road to 500 kWp | 315 kWp, needs other roofs or car-park canopies |

*Table 1: Summary of Energy Audit Finding & Recommended PV Solar Installation*

### Recommendation

Phase 1: Proceed with the 185.4 kWp mezzanine roof system.

Phase 2: Proceed with 250.4 kWp the car park

## 2. Site and Design Basis

### 2.1 Site and load

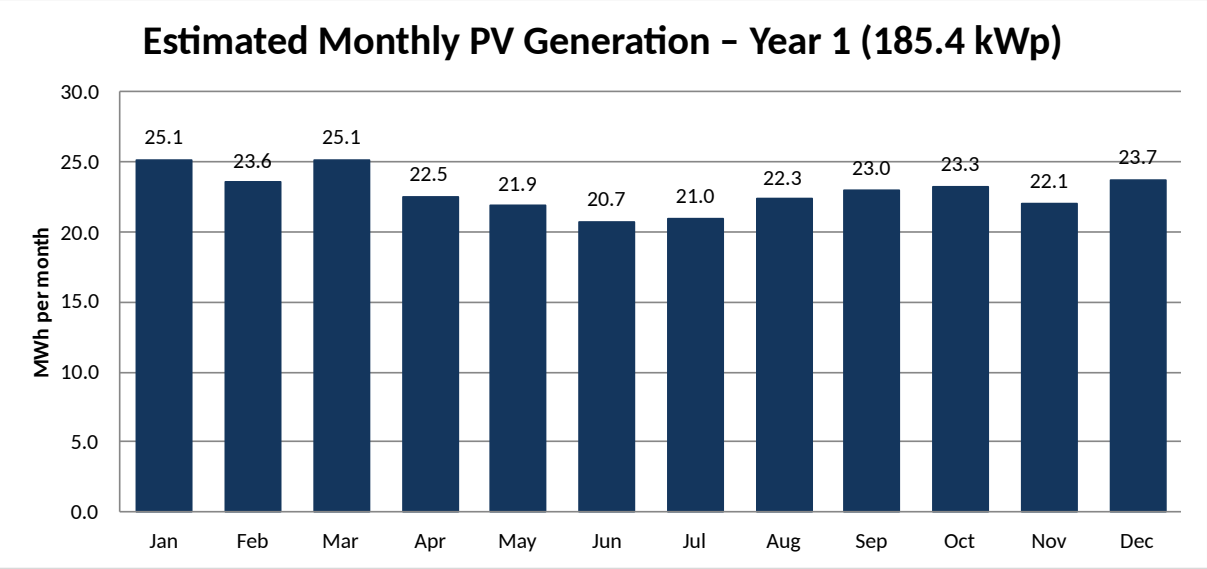
| | |
|---|---|
| **Facility** | Marriott Hotel –PV on the mezzanine-level roof |
| **Location** | 0°18'11.34"N, 32°35'13.76"E |
| **Roof** | Flat reinforced-concrete slab, single level, with parapet. |
| **Annual electricity consumption** | 12,000,000 kWh |
| **Average monthly energy consumption** | 1,000,000 kWh |
| **Average / daytime demand** | |
| **Grid connection** | Behind-the-meter, LV main switchboard |
| **Energy storage** | None |




*Figure 1: PV module installation on the mezzanine roof ( AI Generated )*

## 3. Estimation of PV Solar Generation

- Year-1 generation 274,362 kWh, On Average 22,864 kWh/month and daily 752 kWh/day
- 25-year generation 6,477 MWh*
- share of hotel uses 100 %, 2.3 % of annual consumption




*1.0 % degradation in year 1, then 0.4 %/yr

Note: Output is highest from January to March and lowest in June–July, when cloud cover in Kampala is heaviest.
The difference between the best and worst months is only about 20 %. The hotel load is higher than the PV output in every hour, so all this energy is used on site.

## 4. Solar Impact on Time-of-Use Tariff

The hotel is billed on a time-of-use (TOU) tariff, so each kWh is worth a different amount depending on when it is used.

| Tariff period | Hours | Indicative rate (USD/kWh) |
|---|---|---|
| Off-peak | 00:00–06:00 | 0.054 |
| Shoulder | 06:00–18:00 | 0.070 |
| Peak | 18:00–24:00 | 0.086 |

*Table 2:Bands per uganda ERA TOU structure; rates indicative (USD/kWh)*

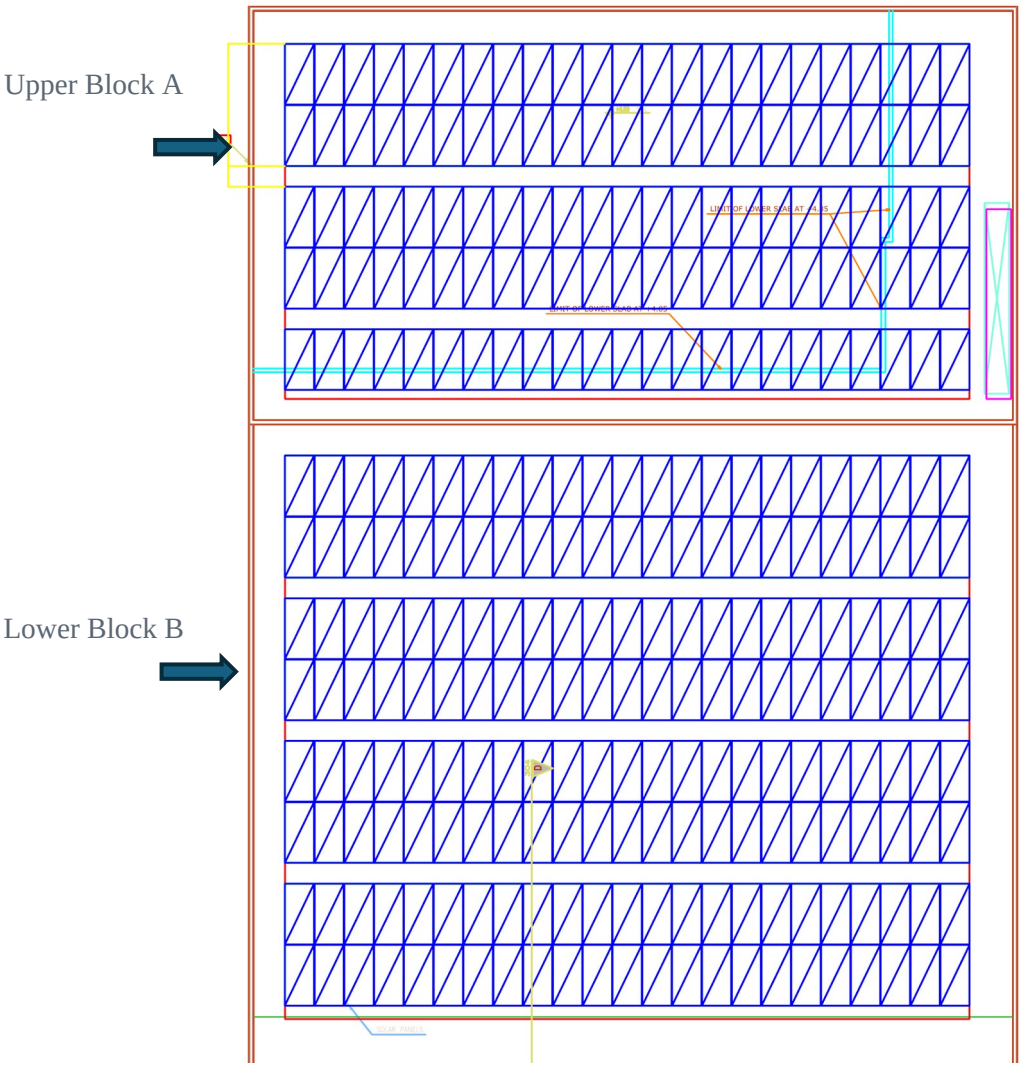
| Period | Hotel use (kWh/yr) | Solar (MWh/yr) | Share of solar | Grid after solar (kWh/yr) | Reduction | Cost before (USD/yr) | Cost after (USD/yr) | Saving (USD/yr) |
|---|---|---|---|---|---|---|---|---|
| Off-peak | 2,090 | 0.0 | 0.0% | 2,090 | 0.00% | 113,648 | 113,648 | 0 |
| Shoulder | 6,954 | 270.3 | 98.5% | 6,684 | 3.89% | 486,965 | 468,030 | 18,934 |
| Peak | 2,956 | 4.0 | 1.5% | 2,952 | 0.14% | 254,079 | 253,739 | 340 |
| **Total** | **12,000** | **274.4** | **100.0%** | **11,726** | **2.29%** | **854,692** | **835,418** | **19,275** |

*Table 3 : Energy reduction percentage & cost saving per year*

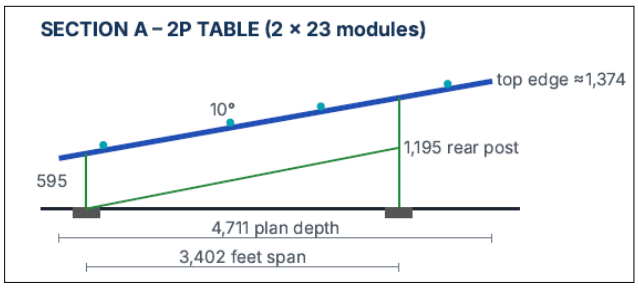
*Forecasted Energy demand for year assumed by september energy bill*12

## 5. PV Array Layout

Each row holds 23 modules on Enerack R80 rails 26,630 mm long, supported by 11 frames per table at 2,600 mm centers. The upper block carries two 2P tables and one 1P table; the lower block carries four 2P tables.




| Zone | Tables | Rows | Modules | kWp |
|---|---|---|---|---|
| Upper block A | 2 × 2P + 1 × 1P | 5 | 115 | 71.3 |
| Lower block B | 4 × 2P | 8 | 184 | 114.1 |
| **Total** | **7 tables** | **13** | **299** | **185.4** |



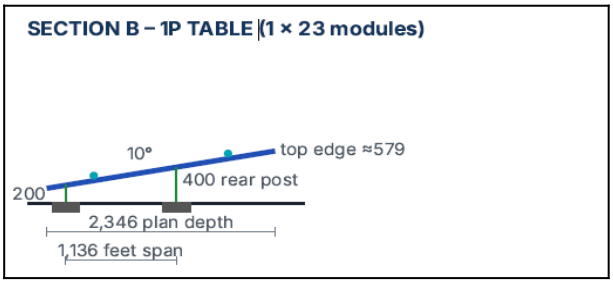




*Figure 4 : PV Layout and Table Structure details,*

## 6. Electrical Design

### 6.1 Configuration

| Item | Inverter INV-01 | Inverter INV-02 | Total |
|---|---|---|---|
| Rated AC power | 80 kW | 80 kW | 160 kW |
| Strings | 8 × 19 modules | 3 × 19 + 5 × 18 modules | 16 strings |
| Modules | 152 | 147 | 299 |
| DC capacity | 94.2 kWp | 91.1 kWp | 185.4 kWp |
| DC/AC ratio | 1.18 | 1.14 | 1.16 |

*Table 5: System Configuration & Inverter Capacity of each block*

### 6.2 Single-line arrangement

1. PV strings: 6 mm² double-insulated PV cable through the cable trays, via a DC isolator and a Type II DC SPD for every string, to the inverters on the roof in a shaded, ventilated location.
2. Inverters (400 V, 3-phase) → AC combiner panel with 2 × 160 A MCCB incomers, Type II AC SPD and 400 A roof isolator.
3. AC combiner panel → 4C × 240 mm² Cu XLPE/SWA → existing Masterpact NW10 H1 (1000 A) solar incomer on the main LV switchboard.
4. Earthing: panel-to-panel bonding of the module frames, connected to the building earth and integrated with the lightning protection system.

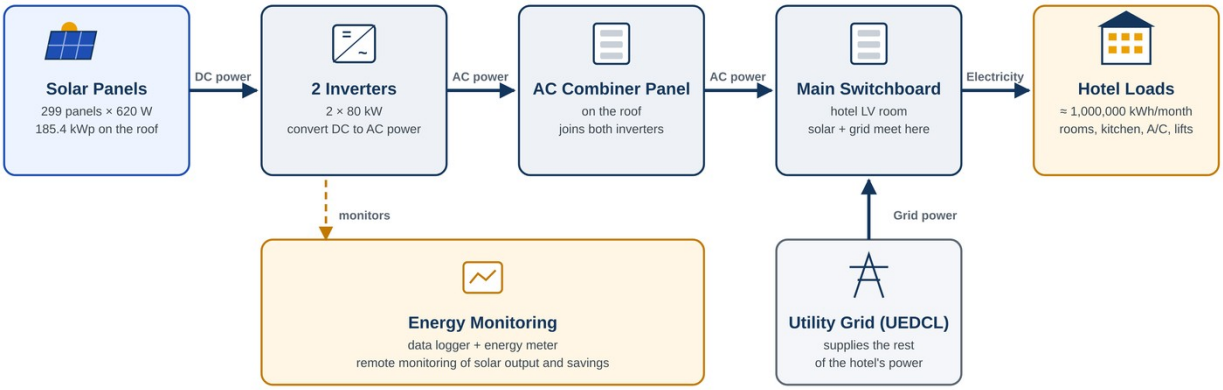



*Figure 5 : Roof Top PV System Block Diagram*

| Component | Description | Size / Quantity |
|---|---|---|
| **Solar panels** | Generate DC electricity | 299 × 620 W = 185.4 kWp |
| **Inverters** | Convert DC power to AC power | 2 × 80 kW |
| **AC combiner panel** | Joins both inverter outputs on the roof | 1 panel, 400 A |
| **Main switchboard** | Where solar and grid power meet (hotel LV room) | Existing, 1000 A |
| **Utility grid (UEDCL)** | Supplies the rest of the hotel's electricity | Existing supply |
| **Expected generation** | Solar energy used by the hotel each year | 274,000 kWh/yr (22,900 kWh/month) |

*Table 6 each item and in the block and Size/Quantity*

## 7. Cost Estimate (Bill of Quantities)

| No. | Description | Qty | Rate (USD) | Amount (USD) |
|---|---|---|---|---|
| 1 | Supply of PV modules, 620 Wp (299 Nos Modules) | 185.4 kWp | 0.13 | 24,099 |
| 2 | Supply of String inverters, 80 kW | 2 nos | 4,000 | 8,000 |
| 3 | Supply of Mounting structure per Wp | 185.4 kWp | 0.095 | 17,634 |
| 4 | Supply of DC system | Lot | USD 0.029/Wp | 5,376 |
| 5 | Supply of AC system | Lot | Lump sum | 6,245 |
| 6 | Earthing, equipotential bonding & lightning protection integration | Lot | Lump sum | 1,847 |
| 7 | Installation, testing & commissioning | Lot | Lump sum | 5,000 |
| 8 | Transport, logistics & customs clearing | Lot | Lump sum | 8,650 |
| | Sub-total | | | 76,851 |
| | **Contingency** | | 7 % | **5,380** |
| | **Total budget EPC cost (excluding taxes)** | | | **82,231** |
| | **Profit margin** | | 25 % | **20,558** |
| | **Total Project Value (excluding taxes)** | | | **102,789** |
| | Specific cost | | | 0.554 USD/Wp |

*Table 7: Project Cost Estimation*

## 8. Financial Analysis

The solar system needs a total investment of USD 102,789 (total project value, excluding taxes). At an avoided grid tariff of USD 0.07/kWh, rising 2% a year, it saves about USD 19,200 in its first year. The investment pays back in about 5.2 years. Over 25 years the project gives an IRR of 19.8% and an NPV of USD 92,606 at a 10% discount rate, with total savings of about USD 579,000.

Solar energy costs about USD 0.043/kWh, below the grid tariff, so every unit generated reduces the hotel's electricity cost. No O&M or inverter-replacement cost is included in the cash flow.

| | |
|---|---|
| Total investment (excluding taxes) | USD 102,789 |
| Avoided grid tariff (year 1) / escalation | USD 0.07/kWh / 2 % per year |
| Year-1 savings | USD 19,212 |
| Simple payback | 5.2 years |
| Project IRR (25 years, unlevered) | 19.8 % |
| NPV at 10 % discount rate | USD 92,606 |
| Levelized cost of energy (LCOE) | USD 0.043/kWh |
| 25-year cumulative savings | USD 578,643 |

*Table 8 : Financial Analysis*

## Tariff sensitivity

The project remains viable across a wide tariff range. Each USD 0.01/kWh change in the avoided tariff moves the NPV by about USD 27.9k and the payback by about 0.6 years. The break-even tariff (NPV = 0 at a 10 % discount rate) is about USD 0.037/kWh, roughly 47 % below the current USD 0.07/kWh, so the investment stays profitable even if the effective tariff falls significantly.

| Avoided tariff (USD/kWh) | Payback (yrs) | IRR | NPV 10 % (USD) |
|---|---|---|---|
| 0.070 | 5.2 | 19.8% | 92,543 |
| 0.080 | 4.6 | 22.6% | 120,447 |
| 0.090 | 4.1 | 25.3% | 148,352 |
| 0.100 | 3.7 | 28.0% | 176,256 |
| 0.120 | 3.1 | 33.3% | 232,065 |

*Table 9: Tariff Sensitivity*

## Annual cash flow (selected years, USD)

For an investment of USD 102,789, the system saves about USD 19,200 in its first year, rising with the tariff*. The investment is recovered in about 5.2 years, and over 25 years the hotel's cumulative net cash flow reaches USD 475,854.

| Year | Generation (kWh) | Tariff (USD/kWh) | Savings | Net cash flow | Cumulative |
|---|---|---|---|---|---|
| 0 | | | | (102,789) | (102,789) |
| 1 | 274,362 | 0.070 | 19,212 | 19,212 | (83,577) |
| 2 | 270,532 | 0.071 | 19,322 | 19,322 | (64,255) |
| 3 | 269,450 | 0.073 | 19,630 | 19,630 | (44,625) |
| 4 | 268,372 | 0.074 | 19,942 | 19,942 | (24,683) |
| 5 | 267,299 | 0.076 | 20,260 | 20,260 | (4,423) |
| **6** | **266,230** | **0.077** | **20,582** | **20,582** | **16,160** |
| **7** | **265,165** | **0.079** | **20,910** | **20,910** | **37,070** |
| **8** | **264,104** | **0.080** | **21,243** | **21,243** | **58,313** |
| **9** | **263,048** | **0.082** | **21,581** | **21,581** | **79,894** |
| **10** | **261,996** | **0.084** | **21,925** | **21,925** | **101,819** |
| **12** | **259,904** | **0.087** | **22,628** | **22,628** | **146,721** |
| **15** | **256,797** | **0.092** | **23,726** | **23,726** | **216,790** |
| **18** | **253,728** | **0.098** | **24,878** | **24,878** | **290,260** |
| **20** | **251,702** | **0.102** | **25,676** | **25,676** | **341,210** |
| **25** | **246,708** | **0.113** | **27,786** | **27,786** | **475,854** |

*Table 10: Annual Cash Flow*

**Grid price saved for each kWh of solar. Starts at USD 0.07 and rises 2% a year to about USD 0.113 by year 25.*

## 9. Expansion Options

The target of about 500 kWp exceeds what the architect's solar zone on the mezzanine roof can hold. Because the hotel load is around ten times the PV output, every additional kWp would also be fully self-consumed, so expansion can be added in stages without a battery.

| Option | Capacity | Year-1 energy | Budget cost (USD) | Remarks |
|---|---|---|---|---|
| Phase 1 – mezzanine roof (this report) | 185.4 kWp | 274 MWh | 102,789 | Ready for detailed design |
| Phase 2 – car-park solar canopies | 250 kWp | 370 MWh | 187,500 | Adds shaded parking; steel canopy cost |
| Indicative total | 435 kWp | 644 MWh | 290,289 | 5.4 % of hotel consumption |

## 10. Assumptions, Exclusions and Next Steps

### Key assumptions

- The hotel load is about 1,000,000 kWh per month with a continuous daytime demand well above 160 kW.
- The avoided tariff is USD 0.07/kWh escalating at 2 %/yr, to be confirmed from the latest utility bills.
- Prices are 2026 East African market levels in USD, excluding taxes.

### Exclusions

- Battery storage, diesel-generator synchronisation and export metering.
- Roof waterproofing renewal, structural strengthening, and civil works to the LV room.
- Utility network upgrades, permit fees and financing costs.

Annexes

1.Single Line Diagram
2.Layout Drawing
3.Data Sheets of Inverter
4.Data Sheet of Solar Module
5.Stuctrural Drawing
6.Quotation